Heat-rejection viability and source water use by Indian climate zone

cell value = WUE_{source} in L/kWh at grid EWIF = 2.30 · ● viable ◐ marginal ○ poor

	Warm & humid Chennai · Mumbai · Vizag *≈70% of capacity* *PUE band from Table II*	Composite Delhi–NCR · Gr. Noida *fastest growing* *PUE band from Table II*	Temperate Bengaluru · Pune *mature market* *PUE band from Table II*
A1 Chilled water + open cooling tower **site water HIGH**	● 5.13	● 5.07	● 4.97
A2 Direct / indirect evaporative **site water HIGH**	○ 5.42	◐ 5.30	● 5.19
A3 Air-cooled chiller, fully dry **site water ≈ 0**	◐ 4.37	● 4.02	● 3.68
A4 Dry cooler + adiabatic assist **site water seasonal**	◐ 4.43	● 4.20	● 4.01
A5 Warm-water DTC + dry cooler **site water ≈ 0**	● 3.38	● 3.15	● 2.92
A6 Immersion + dry cooler **site water ≈ 0**	● 3.20	● 3.02	● 2.79

Δ 0.99 L/kWh

described by all four Indian colocation groups for new or retrofitted capacity

also sold as "water-free"; with A6 the lowest source WUE in every zone

blue outline = best class in that zone

hot & dry zone omitted: evaporative performs well thermally there but water stress is worst, and its PUE parameters are not derived in this study

WUE_{source} L/kWh 2.7 — 4.2 — 5.6 *worse →*